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NEET · BIOLOGY · CHAPTER-WISE PYQS

Molecular Basis of Inheritance

NEET Biology · Previous Year Questions with Answer Key

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QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	253 questions
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Answer key	at the end

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MCQ (Single Correct Answer) 253 questions

Q1. RE-NEET 2026

Which of the following statements about lac-operon is correct?

(A) Galactose can act as an inducer of lac operon

(B) Gene i is constitutively expressed

(C) Lactose activates repressor to bind to the operator

(D) Genes i, z, y and a share single common promoter

Open solution

Q2. RE-NEET 2026

Which of the following enzymes synthesizes precursor mRNA?

(A) DNA polymerase

(B) RNA polymerase I

(C) RNA polymerase II

(D) RNA polymerase III

Open solution

Q3. NEET 2026

In the lac operon, the z gene codes for

(A) permease

(B) transacetylase

(C) beta-galactosidase

(D) the repressor of lac operon

Open solution

Q4. NEET 2026

Which of the following statements are correct with reference to a transcription unit?

A. A transcription unit in DNA is defined primarily by three regions : promoter, structural gene and terminator.

B. The promoter is said to be located towards the 5'-end of the structural gene.

C. The promoter is a DNA sequence that provides binding site for RNA polymerase.

D. The promoter defines the template and coding strands.

E. The terminator is located towards the $3'$-end of the coding strand and it defines the end of the process of transcription.

Choose the correct answer from the options given below:

(A) A, B, C and D only

(B) A, C, D and E only

(C) B, C, D and E only

(D) A, B, C, D and E

Open solution

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Q5. NEET 2026

Arrange the following steps of DNA fingerprinting in a correct sequence.

A. Isolation of DNA and its digestion by restriction endonucleases.

B. Hybridisation using a labelled VNTR probe.

C. Transferring of separated DNA fragments to synthetic membranes.

D. Detection of hybridised DNA fragments by autoradiography.

E. Separation of DNA fragments by electrophoresis.

Choose the correct answer from the options given below :

(A) $\mathrm{A}, \mathrm{B}, \mathrm{D}, \mathrm{C}, \mathrm{E}$

(B) $\mathrm{A}, \mathrm{D}, \mathrm{B}, \mathrm{E}, \mathrm{C}$

(C) A, E, C, B, D

(D) $\mathrm{A}, \mathrm{E}, \mathrm{B}, \mathrm{C}, \mathrm{D}$

Open solution

Q6. NEET 2026

Which of the following statements are correct with reference to packaging of DNA helix ?

A. Histones are organized to form a unit of eight molecules called histone octamer.

B. Histones are negatively charged basic proteins.

C. Histones are rich in the basic amino acid residues - lysine and arginine.

D. The positively charged DNA is wrapped around the histone octamer to form nucleosome.

E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the correct answer from the options given below :

(A) A, C and E only (B) B, D and E only (C) C, D and E only (D) A, B and D only

Open solution

Q7. NEET 2026

The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ____

(A) GUG (B) AUG (C) GAG (D) CAG

Open solution

Q8. NEET 2025

Which chromosome in the human genome has the highest number of genes?

(A) Chromosome 1 (B) Chromosome 10 (C) Chromosome X (D) Chromosome Y

Open solution

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Q9. NEET 2025

Open solution

Match List-I with List-II

	List - I		List - II
A	Alfred Hershey and Martha Chase	I	Streptococcus pneumoniae
B	Euchromatin	II	Densely packed and dark-stained
C	Frederick Griffith	III	Loosely packed and light-stained
D	Heterochromatin	IV	DNA as genetic material confirmation

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-II, C-IV, D-I
(C) A-II, B-IV, C-I, D-III
(D) A-IV, B-II, C-I, D-III

Q10. NEET 2025

Open solution

Which of the following are the post-transcriptional events in an eukaryotic cell?

A. Transport of pre-mRNA to cytoplasm prior to splicing.

B. Removal of introns and joining of exons.

C. Addition of methyl group at 5' end of hnRNA.

D. Addition of adenine residues at 3' end of hnRNA.

E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below :

(A) B, C, E only **(B)** C, D, E only **(C)** A, B, C only **(D)** B, C, D only

Q11. NEET 2025

Open solution

Given below are two statements:

Statement I: In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect
(B) Statement I is incorrect but statement II is correct
(C) Both statement I and statement II are correct
(D) Both statement I and statement II are incorrect

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Q12. NEET 2025

Open solution

Given below are two statements :

Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but statement II is correct

(C) Both statement I and statement II are correct

(D) Both statement I and statement II are incorrect

Q13. NEET 2025

Open solution

Who proposed that the genetic code for amino acids should be made up of three nucleotides?

(A) Jacque Monod (B) Franklin Stahl (C) George Gamow (D) Francis Crick

Q14. NEET 2025

Open solution

Histones are enriched with -

(A) Phenylalanine & Leucine

(B) Phenylalanine & Arginine

(C) Lysine & Arginine

(D) Leucine & Lysine

Q15. NEET 2025

Open solution

Which factor is important for termination of transcription?

(A) ρ (rho) (B) γ (gamma) (C) α (alpha) (D) σ (sigma)

Q16. NEET 2024 (Re-Examination)

Open solution

In a chromosome, there is a specific DNA sequence, responsible for initiating replication. It is called as:

(A) Recognition sequence

(B) Cloning site

(C) Restriction site

(D) ori site

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Q17. NEET 2024 (Re-Examination)

Open solution

Given below are two statements regarding RNA polymerase in prokaryotes.

Statement I : In prokaryotes, RNA polymerase is capable of catalysing the process of elongation during transcription.

Statement II : RNA polymerase associate transiently with 'Rho' factor to initiate transcription.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q18. NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: In eukaryotes there are three RNA polymerases in the nucleus in addition to the RNA polymerase found in the organelles.

Statement II: All the three RNA polymerases in eukaryotic nucleus have different roles.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q19. NEET 2024 (Re-Examination)

Open solution

Given below are two statements :

Statement I: In the lac operon, the z gene codes for beta-galactosidase which is primarily responsible for the hydrolysis of lactose into galactose and glucose.

Statement II: In addition to lactose, glucose or galactose can also induce lac operon.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

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Q20. NEET 2024

The lactose present in the growth medium of bacteria is transported to the cell by the action of

(A) Beta-galactosidase
(B) Acetylase
(C) Permease
(D) Polymerase

Open solution

Q21. NEET 2024

A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;

(A) Repressor, Operator gene, Structural gene
(B) Structural gene, Transposons, Operator gene
(C) Inducer, Repressor, Structural gene
(D) Promotor, Structural gene, Terminator

Open solution

Q22. NEET 2024

Which of the following statement is correct regarding the process of replication in E.coli?

(A) 'The DNA dependent DNA polymerase catalyses polymerization in one direction that is $3' \rightarrow 5'$
(B) The DNA dependent RNA polymerase catalyses polymerization in one direction, that is $5' \rightarrow 3'$
(C) The DNA dependent DNA polymerase catalyses polymerization in 5' $\rightarrow$ 3' as well as $3' \rightarrow 5'$ direction
(D) The DNA dependent DNA polymerase catalyses polymerization in 5' $\rightarrow$ 3' direction

Open solution

Q23. NEET 2024

Match List I with List II

	List - I		List - II
A.	Frederick Griffith	I.	Genetic code
B.	Francois Jacob & Jacque Monod	II.	Semi-conservative mode of DNA replication
C.	Har Gobind Khorana	III.	Tranformation
D.	Meselson & Stahl	IV.	Lac operon

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV
(B) A-III, B-IV, C-I, D-II
(C) A-II, B-III, C-IV, D-I
(D) A-IV, B-II, C-II, D-III

Open solution

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Q24. NEET 2024

Which one is the correct product of DNA dependent RNA polymerase to the given template?

3'TACATGGCAAATATCCATTCA5'

Open solution

(A) 5'AUGUACCGUUUAUAGGUAAGU3'
(B) 5'AUGUAAAGUUUAUAGGUAAGU3'
(C) 5'AUGUACCGUUUAUAGGGAAGU3'
(D) 5'ATGTACCGTTTATAGGTAAGT3'

Q25. NEET 2024

Match List I with List II:

Open solution

	List - I		List - II
A.	RNA polymerase III	I.	snRNPs
B.	Termination of transcription	II.	Promotor
C.	Splicing of Exons	III.	Rho factor
D.	TATA box	IV.	SnRNAs, tRNA

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III
(B) A-III, B-II, C-IV, D-I
(C) A-III, B-IV, C-I, D-II
(D) A-IV, B-III, C-I, D-II

Q26. NEET 2023 Manipur

The last chromosome sequenced in Human Genome Project was:

Open solution

(A) Chromosome 6 (B) Chromosome 1 (C) Chromosome 22 (D) Chromosome 14

Q27. NEET 2023 Manipur

Name the component that binds to the operator region of an operon and prevents RNA polymerase from transcribing the operon.

Open solution

(A) Promotor
(B) Regulator protein
(C) Repressor protein
(D) Inducer

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Q28. NEET 2023 Manipur

Open solution

Given below are two statements :

Statement I : The process of copying genetic information from one strand of the DNA into RNA is termed as transcription.

Statement II : A transcription unit in DNA is defined primarily by the three regions in the DNA i.e., a promotor, the structural gene and a terminator.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q29. NEET 2023 Manipur

Open solution

Which scientist conducted an experiment with ^{32}P and ^{35}S labelled phages for demonstrating that DNA is the genetic material?

(A) james D. Watson and F.H. C. Crick

(B) A. D Hershey and M.J. Chase

(C) F. Griffith

(D) O.T. Avery, C.M. MacLeod and M. McCarty

Q30. NEET 2023 Manipur

Open solution

Given below are two statements :

Statement I : RNA being unstable, mutate at a faster rate.

Statement II : RNA can directly code for synthesis of proteins hence can easily express the characters.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is correct but Statement II is false

(B) Statement I is incorrect but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q31. NEET 2023 Manipur

Open solution

Which one of the following acts as an inducer for lac operon ?

(A) Sucrose

(B) Lactose

(C) Glucose

(D) Galactose

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Q32. NEET 2023 Manipur

With reference to Hershey and Chase experiments. Select the correct statements.

(A) Viruses grown in the presence of radioactive phosphorus contained radioactive DNA.

(B) Viruses grown on radioactive sulphur contained radioactive proteins.

(C) Viruses grown on radioactive phosphorus contained radioactive protein.

(D) Viruses grown on radioactive sulphur contained radioactive DNA.

(E) Viruses grown on radioactive protein contained radioactive DNA.

Choose the most appropriate answer from the options given below :

(A) (D) and (E) only **(B)** (A) and (B) only **(C)** (A) and (C) only **(D)** (B) and (D) only

Open solution

Q33. NEET 2023 Manipur

The salient features of genetic code are :

(A) The code is palindromic

(B) UGA act as initiator codon

(C) The code is unambiguous and specific

(D) The code is nearly universal

Choose the most appropriate answer from the options given below :

(A) (A) and (D) only **(B)** (B) and (C) only **(C)** (A) and (B) only **(D)** (C) and (D) only

Open solution

Q34. NEET 2023

Unequivocal proof that DNA is the genetic material was first proposed by

(A) Alfred Hershey and Martha Chase **(B)** Avery, Macleoid and McCarthy

(C) Wilkins and Franklin **(D)** Frederick Griffith

Open solution

Q35. NEET 2023

What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

(A) Transcription of tRNA, 5S rRNA and snRNA

(B) Transcription of precursor of mRNA

(C) Transcription of only snRNAs

(D) Transcription of rRNAs (28S, 18S and 5.8S)

Open solution

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Q36. NEET 2023

Expressed Sequence Tags (ESTs) refers to

(A) All genes that are expressed as proteins.

(B) All genes whether expressed or unexpressed.

(C) Certain important expressed genes.

(D) All genes that are expressed as RNA.

Open solution

Q37. NEET 2023

Match List I with List II.

	List I		List II
(A)	Gene 'a'	(I)	β-galactosidase
(B)	Gene 'y'	(II)	Transacetylase
(C)	Gene 'i'	(III)	Permease
(D)	Gene 'z'	(IV)	Repressor protein

Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I

(B) A-III, B-IV, C-I, D-II

(C) A-III, B-I, C-IV, D-II

(D) A-II, B-I, C-IV, D-III

Open solution

Q38. NEET 2023

Given below are two statements:

Statement I: RNA mutates at a faster rate.

Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false.

(B) Statement I is true but Statement II is false.

(C) Statement I is false but Statement II is true.

(D) Both Statement I and Statement II are true.

Open solution

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Q39. NEET 2023

Open solution

Given below are two statements:

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false.

(B) Statement I is correct but Statement II is false.

(C) Statement I is incorrect but Statement II is true.

(D) Both Statement I and Statement II are true.

Q40. NEET 2023

Open solution

Which one of the following is the sequence on corresponding coding strand, if the sequence on mRNA formed is as follows 5'AUCGAUCGAUCGAUCGAUCGAUCG 3'?

(A) 3' UAGCUAGCUAGCUAGCUAGCUAGC 5'

(B) 5' ATCGATCGATCGATCGATCGATCG 3'

(C) 3' ATCGATCGATCGATCGATCGATCG 5'

(D) 5' UAGCUAGCUAGCUAGCUAGCUAGC 3'

Q41. NEET 2022 Phase 2

Open solution

In lac operon, z gene codes for

(A) Transacetylase

(B) β-galactosidase

(C) Permease

(D) Repressor

Q42. NEET 2022 Phase 2

Open solution

Given below are two statements

Statement I : DNA polymerases catalyse polymerisation only in one direction, that is 5' → 3'.

Statement II : During replication of DNA, on one strand the replication is continuous while on other strand it is discontinuous.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

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Q43. NEET 2022 Phase 2

Open solution

Match List - I with List - II.

	List - I		List - II
(a)	In lac operon i gene codes for	(i)	transacetylase
(b)	In lac operon z gene codes for	(ii)	permease
(c)	In lac operon y gene codes for	(iii)	β-galactosidase
(d)	In lac operon a gene codes for	(iv)	Repressor

Choose the correct answer from the options given below

(A) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

(B) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(C) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(D) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Q44. NEET 2022 Phase 2

Open solution

Match List-I with List-II:

	List - I		List - II
(a)	Bacteriophage $\phi \times 174$	(i)	48502 base pairs
(b)	Bacteriophage lambda	(ii)	5386 nucleotides
(c)	Escherichia coli	(iii)	3.3×10^9 base pairs
(d)	Haploid content of human DNA	(iv)	4.6×10^6 base pairs

Choose the correct answer from the options given below:

(A) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

(B) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(C) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Q45. NEET 2022 Phase 2

Open solution

If DNA contained sulfur instead of phosphorus and proteins contained phosphorus instead of sulfur, what would have been the outcome of Hershey and Chase experiment?

(A) Radioactive phosphorus in bacterial cells

(B) No radioactive sulfur in bacterial cells

(C) Both radioactive sulfur and phosphorus in bacterial cells

(D) Radioactive sulfur in bacterial cells

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Q46. NEET 2022 Phase 2

Against the codon 5' UAC 3', what would be the sequence of anticodon on tRNA?

(A) 5' GUA 3' (B) 5' AUG 3' (C) 5' ATG 3' (D) 5' GTA 3'

Open solution

Q47. NEET 2022 Phase 2

If A and C make 30% and 20% of DNA, respectively, what will be the percentage composition of T and G?

(A) T : 20%, G : 20% (B) T : 20%, G : 30% (C) T : 30%, G : 20% (D) T : 30%, G : 30%

Open solution

Q48. NEET 2022 Phase 1

The process of translation of mRNA to proteins begins as soon as:

(A) The small subunit of ribosome encounters mRNA

(B) The larger subunit of ribosome encounters mRNA

(C) Both the subunits join together to bind with mRNA

(D) The tRNA is activated and the larger subunit of ribosome encounters mRNA

Open solution

Q49. NEET 2022 Phase 1

DNA polymorphism forms the basis of:

(A) Genetic mapping

(B) DNA finger printing

(C) Both genetic mapping and DNA finger printing

(D) Translation

Open solution

Q50. NEET 2022 Phase 1

Read the following statements and choose the set of correct statements:

(a) Euchromatin is loosely packed chromatin

(b) Heterochromatin is transcriptionally active

(c) Histone octamer is wrapped by negatively charged DNA in nucleosome

(d) Histones are rich in lysine and arginine

(e) A typical nucleosome contains 400 bp of DNA helix

Choose the correct answer from the options given below :

(A) (b), (d), (e) only (B) (a), (c), (d) only

(C) (b), (e) only (D) (a), (c), (e) only

Open solution

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Q51. NEET 2022 Phase 1

If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as :

(A) Sequence annotation
(B) Gene mapping
(C) Expressed sequence tags
(D) Bioinformatics

Open solution

Q52. NEET 2022 Phase 1

If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs?

(A) 3.3×10^9 bp
(B) 6.6×10^9 bp
(C) 3.3×10^6 bp
(D) 6.6×10^6 bp

Open solution

Q53. NEET 2022 Phase 1

In an E. Coli strain i gene gets mutated and its product can not bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome?

(A) Only z gene will get transcribed
(B) z, y, a genes will be transcribed
(C) z, y, a genes will not be translated
(D) RNA polymerase will bind the promoter region

Open solution

Q54. NEET 2022 Phase 1

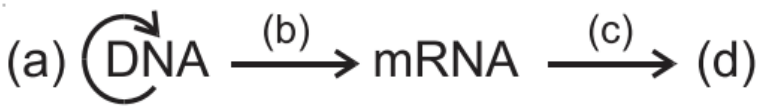
Ten E.coli cells with ^{15}N - dsDNA are incubated in medium containing ^{14}N nucleotide. After 60 minutes, how many E.coli cells will have DNA totally free from ^{15}N?

(A) 20 cells
(B) 40 cells
(C) 60 cells
(D) 80 cells

Open solution

Q55. NEET 2021

Complete the flow chart on central dogma.



(A) (a) - Transduction; (b) - Translation; (c) - Replication; (d) - Protein
(B) (a) - Replication; (b) - Transcription; (c) - Transduction; (d) - Protein
(C) (a) - Translation; (b) - Replication; (c) - Transcription; (d) - Transduction
(D) (a) - Replication; (b) - Transcription; (c) - Translation; (d) - Protein

Open solution

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Q56. NEET 2021

What is the role of RNA polymerase III in the process of transcription in eukaryotes?

(A) Transcribes only snRNAs
(B) Transcribes rRNAs (28S, 18S and 5.8S)
(C) Transcribes tRNA, 5s rRNA and snRNA
(D) Transcribes precursor of mRNA

Open solution

Q57. NEET 2021

DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as

(A) Polymorphic DNA
(B) Satellite DNA
(C) Repetitive DNA
(D) Single nucleotides

Open solution

Q58. NEET 2021

Identify the correct statement.

(A) Split gene arrangement is characteristic of prokaryotes
(B) In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA
(C) RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria
(D) The coding strand in a transcription unit is copied to an mRNA

Open solution

Q59. NEET 2021

If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine and Cytosine in it?

(A) T : 20 ; G : 25 ; C : 25
(B) T : 20 ; G : 30 ; C : 20
(C) T : 20 ; G : 20 ; C : 30
(D) T : 30 ; G : 20 ; C : 20

Open solution

Q60. NEET 2021

Which of the following RNAs is not required for the synthesis of protein?

(A) siRNA (B) mRNA (C) tRNA (D) rRNA

Open solution

Q61. NEET 2021

Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes?

(A) DNase
(B) DNA dependent DNA polymerase
(C) DNA dependent RNA polymerase
(D) DNA Ligase

Open solution

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Q62. NEET 2021

Which one of the following statements about histones is wrong?

(A) Histones carry positive charge in the side chain

(B) Histones are organized to form a unit of 8 molecules

(C) The pH of histones is slightly acidic

(D) Histones are rich in amino acids - Lysine and Arginine

Open solution

Q63. NEET 2021

Statement I : The codon 'AUG' codes for methionine and phenylalanine.

Statement II : 'AAA' and 'AAG' both codons code for the amino acid lysine.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is incorrect but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is correct but Statement II is false

Open solution

Q64. NEET 2020 Phase 2

In the polynucleotide chain of DNA, a nitrogenous base is linked to the $-\mathrm{OH}$ of :

(A) $1'\mathrm{C}$ pentose sugar

(B) $2'\mathrm{C}$ pentose sugar

(C) $3'\mathrm{C}$ pentose sugar

(D) $5'\mathrm{C}$ pentose sugar

Open solution

Q65. NEET 2020 Phase 2

The term 'Nuclein' for the genetic material was used by :

(A) Mendel (B) Franklin (C) Meischer (D) Chargaff

Open solution

Q66. NEET 2020 Phase 2

Which is the basis of genetic mapping of human genome as well as DNA finger printing ?

(A) Polymorphism in RNA sequence

(B) Polymorphism in DNA sequence

(C) Single nucleotide polymorphism

(D) Polymorphism in hnRNA sequence

Open solution

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Q67. NEET 2020 Phase 2

E. Coli has only 4.6×10^6 base pairs and completes the process of replication within 18 minutes; then the average rate of polymerisation is approximately -

(A) 1000 base pairs/second
(B) 2000 base pairs/second
(C) 3000 base pairs/second
(D) 4000 base pairs/second

Open solution

Q68. NEET 2020 Phase 1

If the distance between two consecutive base pairs is 0.34 nm and the total number of base pairs of a DNA double helix in a typical mammalian cell is 6.6×10^9 bp, then the length of the DNA is approximately

(A) 2.7 meters
(B) 2.0 meters
(C) 2.5 meters
(D) 2.2 meters

Open solution

Q69. NEET 2020 Phase 1

Which of the following statements is correct?

(A) Adenine does not pair with thymine
(B) Adenine pairs with thymine through one H-bond
(C) Adenine pairs with thymine through two H-bonds
(D) Adenine pairs with thymine through three H-bonds

Open solution

Q70. NEET 2020 Phase 1

The first phase of translation is

(A) Recognition of an anti-codon
(B) Recognition of DNA molecule
(C) Binding of mRNA to ribosome
(D) Aminoacylation of tRNA

Open solution

Q71. NEET 2020 Phase 1

Name the enzyme that facilitates opening of DNA helix during transcription.

(A) DNA polymerase
(B) RNA polymerase
(C) DNA helicase
(D) DNA ligase

Open solution

Q72. NEET 2019 (Odisha)

From the following, identify the **correct combination** of salient features of Genetic Code -

(A) Degenerate, Non-overlapping, Non-ambiguous
(B) Universal, Non-ambiguous, Overlapping
(C) Degenerate, Overlapping, Commaless
(D) Universal, Ambiguous, Degenerate

Open solution

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Q73. NEET 2019 (Odisha)

Which scientist experimentally proved that DNA is the sole genetic material in bacteriophage?

(A) Jacob and Monod
(B) Beadle and Tautum
(C) Messelson and Stahl
(D) Hershey and Chase

Open solution

Q74. NEET 2019 (Odisha)

In the process of transcription in Eukaryotes, the RNA polymerase I transcribes -

(A) Precursor of mRNA, hnRNA
(B) mRNA with additional processing, capping and tailing
(C) tRNA, 5 srRNA and snRNAs
(D) rRNAs – 28 S, 18 S and 5.8 S

Open solution

Q75. NEET 2019 (Odisha)

What initiation and termination factors are involved in transcription in Eukaryotes?

(A) α and σ, respectively
(B) σ and ρ, respectively
(C) α and β, respectively
(D) β and γ, respectively

Open solution

Q76. NEET 2019 (Odisha)

Match the following RNA polymerases with their transcribed products:

(a) RNA polymerase I	(i) tRNA
(b) RNA polymerase II	(ii) rRNA
(c) RNA polymerase III	(iii) hnRNA

Select the correct option from the following:

(A) (a)-(iii), (b)-(ii), (c)-(i)
(B) (a)-(i), (b)-(iii), (c)-(ii)
(C) (a)-(i), (b)-(ii), (c)-(iii)
(D) (a)-(ii), (b)-(iii), (c)-(i)

Open solution

Q77. NEET 2019 (Odisha)

What will be the sequence of mRNA produced by the following stretch of DNA?

3' ATGCATGCATGCATG 5' TEMPLATE STRAND

5' TACGTACGTACGTAC 3' CODING STRAND

(A) 3' AUGCAUGCAUGCAUG 3'
(B) 3' AUGCAUGCAUGCAUG 5'
(C) 5' UACGUACGUACGUAC 3'
(D) 3' UACGUACGUACGUAC 5'

Open solution

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Q78. NEET 2019

Purines found both in DNA and RNA are:

(A) Guanine and cytosine
(B) Cytosine and thymine
(C) Adenine and thymine
(D) Adenine and guanine

Open solution

Q79. NEET 2019

Expressed Sequence Tags (ESTs) refers to :

(A) DNA polymorphism
(B) Novel DNA sequences
(C) Genes expressed as RNA
(D) Polypeptide expression

Open solution

Q80. NEET 2019

Under which of the following conditions will there be no change in the reading frame of following mRNA?

5' AACAGCGGUGCUAUU 3'

(A) Insertion of A and G at 4th and 5th positions respectively
(B) Deletion of GGU from 7th, 8th and 9th positions
(C) Insertion of G at 5th positions
(D) Deletion of G from 5th positions

Open solution

Q81. NEET 2019

Which of the following features of genetic code does allow bacteria to produce human insulin by recombinant DNA technology?

(A) Genetic code is nearly universal
(B) Genetic code is specific
(C) Genetic code is not ambiguous
(D) Genetic code is redundant

Open solution

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Q82. NEET 2019

Match the following genes of the Lac operon with their respective products :

(a) i gene	(i) β-galactosidase
(b) z gene	(ii) Permease
(c) a gene	(iii) Repressor
(d) y gene	(iv) Transacetylase

Select the **correct** option.

(A)

(a)	(b)	(c)	(d)
(iii)	(i)	(iv)	(ii)

(B)

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

(C)

(a)	(b)	(c)	(d)
(i)	(iii)	(ii)	(iv)

(D)

(a)	(b)	(c)	(d)
(iii)	(i)	(ii)	(iv)

Open solution

Q83. NEET 2018

Select the correct statement :

(A) Franklin Stahl coined the term “linkage”.

(B) Punnett square was developed by a British scientist.

(C) Spliceosomes take part in translation.

(D) Transduction was discovered by S. Altman

Open solution

Q84. NEET 2018

The experimental proof for semi-conservative replication of DNA was first shown in a

(A) fungus (B) bacterium (C) plant (D) virus.

Open solution

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Q85. NEET 2018

Select the correct match :

(A) Alec Jeffreys - Streptococcus pneumoniae

(B) Alfred Hershey and Martha Chase - TMV

(C) Matthew Meselson and F. Stahl - Pisum sativum

(D) Francois Jacob and Jacques Monod - Lac operon

Open solution

Q86. NEET 2018

Select the correct match :

(A) Ribozyme - Nucleic acid

(B) $F_2 \times$ Recessive parent - Dihybrid cross

(C) T.H. Morgan - Transduction

(D) G. Mendel - Transformation

Open solution

Q87. NEET 2018

Many ribosomes may associate with a single mRNA to form multiple copies of a polypeptide simultaneously. Such strings of ribosomes are termed as

(A) polysome

(B) polyhedral bodies

(C) plastidome

(D) nucleosome

Open solution

Q88. NEET 2018

AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed mRNA?

(A) AGGUAUCGCAU

(B) UGGTUTCGCAT

(C) ACCUAUGCGAU

(D) UCCAUAGCGUA

Open solution

Q89. NEET 2018

All of the following are part of an operon except

(A) an operator

(B) structural genes

(C) an enhancer

(D) a promoter

Open solution

Q90. NEET 2017

DNA replication in bacteria occurs

(A) within nucleolus

(B) prior to fission

(C) just before transcription

(D) during S phase

Open solution

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Q91. NEET 2017

The final proof for DNA as the genetic material came from the experiments of

(A) Griffith
(B) Hargobind Khorana
(C) Avery, Mcleod and McCarty
(D) Hershey and chase

Open solution

Q92. NEET 2017

If there are 999 bases in an RNA that codes for a protein with 333 amino acids, and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered ?

(A) 1
(B) 11
(C) 333
(D) 33

Open solution

Q93. NEET 2017

During DNA replication, Okazaki fragments are used to elongate

(A) The leading strand away from replication fork
(B) The lagging strand towards replication fork
(C) The lagging strand away from the replication fork
(D) The leading strand towards replication fork

Open solution

Q94. NEET 2017

Which of the following RNAs should be most abundant in animal cell ?

(A) m-RNA
(B) mi-RNA
(C) r-RNA
(D) t-RNA

Open solution

Q95. NEET 2017

Spliceosomes are not found in cells of :

(A) Bacteria
(B) Plants
(C) Animals
(D) Fungi

Open solution

Q96. NEET 2017

The association of histone H_1 with a nucleosome indicates :

(A) DNA replication is occurring
(B) Transcription is occurring
(C) The DNA is condensed into a Chromatin Fibre
(D) The DNA double helix is exposed

Open solution

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Q97. NEET 2016 Phase 2

Taylor conducted the experiments to prove semiconservative mode of chromosome replication on

(A) Vicia faba
(B) Drosophila melanogaster
(C) Vinca rosea
(D) E. coli

Open solution

Q98. NEET 2016 Phase 2

The equivalent of a structural gene is -

(A) cistron
(B) recon
(C) muton
(D) operon

Open solution

Q99. NEET 2016 Phase 2

Which of the following rRNAs acts as structural RNA as well as ribozyme in bacteria ?

(A) 5.8 S rRNA
(B) 23 S rRNA
(C) 18 S rRNA
(D) 5 S rRNA

Open solution

Q100. NEET 2016 Phase 2

A molecule that can act as a genetic material must fulfill the traits given below, except

(A) it should be able to express itself in the form of 'Mendelian characters'
(B) it should be unstable structurally and chemically
(C) it should provide the scope for slow changes that are required for evolution
(D) it should be able to generate its replica

Open solution

Q101. NEET 2016 Phase 2

DNA-dependent RNA polymerase catalyzes transcription on one strand of the DNA which is called the

(A) coding strand
(B) antistrand
(C) template strand
(D) alpha strand

Open solution

Q102. NEET 2016 Phase 1

Which one of the following is the starter codon ?

(A) AUG
(B) UAG
(C) UAA
(D) UGA

Open solution

Q103. NEET 2016 Phase 1

Which of the following is required as inducer(s) for the expression of Lac operon?

(A) lactose
(B) lactose and galactose
(C) glucose
(D) galactose

Open solution

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Q104. NEET 2016 Phase 1

A complex of ribosomes attached to a single strand of RNA is known as :

(A) Polypeptide (B) Okazaki fragment (C) Polysome (D) Polymer

Open solution

Q105. NEET 2016 Phase 1

Which of the following is not required for any of the techniques of DNA fingerprinting available at present?

(A) DNA – DNA hybridization
(B) Polymerase chain reaction
(C) Zinc finger analysis
(D) Restriction enzymes

Open solution

Q106. AIPMT 2015

Which one of the following is not applicable to RNA?

(A) Complementary base pairing
(B) 5' phosphoryl and 3' hydroxyl ends
(C) Heterocyclic nitrogenous bases
(D) Chargaff's rule

Open solution

Q107. AIPMT 2015

Balbiani rings are sites of :

(A) Polysaccharide synthesis
(B) Lipid synthesis
(C) RNA and protein synthesis
(D) Nucleotide synthesis

Open solution

Q108. AIPMT 2015

Identify the correct order of organisation of genetic material from largest to smallest :

(A) Chromosome, genome, nucleotide, gene
(B) Genome, chromosome, nucleotide, gene
(C) Genome, chromosome, gene, nucleotide
(D) Chromosome, gene, genome, nucleotide

Open solution

Q109. AIPMT 2015

Satellite DNA is important because it :

(A) Shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.
(B) does not code for proteins and is same in all members of the population.
(C) codes for enzymes needed for DNA replication.
(D) codes for proteins needed in cell cycle.

Open solution

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Q110. AIPMT 2015 Cancelled Paper

Gene regulation governing lactose operon of E.coli that involves the lac I gene product is

(A) negative and inducible because repressor protein prevents transcription.

(B) positive and inducible because it can be induced by lactose

(C) negative and repressible because repressor protein prevents transcription

(D) feedback inhibition because excess of β-galactosidase can switch off transcription

Open solution

Q111. AIPMT 2015 Cancelled Paper

In sea urchin DNA, which is double stranded, 17% of the bases were shown to be cytosine, The percentages of the other three bases expected to be present in this DNA are

(A) G 17%, A 16.5%, T 32.5%

(B) G 17%, A 33%, T 33%

(C) G 8.5%, A 50%, T 24.5%

(D) G 34%, A 24.5%, T 24.5%

Open solution

Q112. AIPMT 2014

Transformation was discovered by :

(A) Griffith

(B) Watson and Crick

(C) Hershey and Chase

(D) Meselson and Stahl

Open solution

Q113. AIPMT 2014

Select the correct option ;

(A) Direction of RNA synthesis 3'–5', Direction of reading of the template DNA strand 5'–3'

(B) Direction of RNA synthesis 5'–3', Direction of reading of the template DNA strand 5'–3'

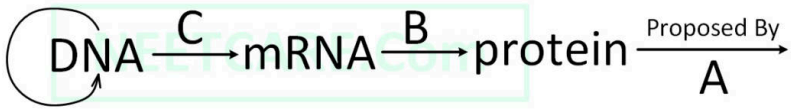
(C) Direction of RNA synthesis 5'–3', Direction of reading of the template DNA strand 3'–5'

(D) Direction of RNA synthesis 3'–5', Direction of reading of the template DNA strand 3'–5'

Open solution

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Q114. NEET 2013 (Karnataka)



Open solution

The figure gives an important concept in the genetic implication of DNA . Fill the blanks A, B and C.

(A) A - Erwin Chargaff
B - translation
C - replication

(B) A - Maurice Wilkins
B - transcription
C - translation

(C) A - Francis Crick
B - translation
C - transcription

(D) A - James Watson
B - replication
C - extension

Q115. NEET 2013 (Karnataka)

Satellite RNA are present in some :

(A) Prions (B) Bacteriophages (C) Plant viruses (D) Viroids

Open solution

Q116. NEET 2013 (Karnataka)

Which of the following is not a property of the genetic code ?

(A) Ambiguous (B) Universal (C) Degeneracy (D) Non-overlapping

Open solution

Q117. NEET 2013 (Karnataka)

Genes of interest can be selected from a genomic library by using :

(A) Restriction enzymes (B) Gene targets
(C) Cloning vectors (D) DNA probes

Open solution

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Q118. NEET 2013 (Karnataka)

In an inducible operon, the genes are :

(A) usually expressed unless a signal turns them "off"

(B) always expressed

(C) never expressed

(D) usually not expressed unless a signal turns them "on"

Open solution

Q119. NEET 2013 (Karnataka)

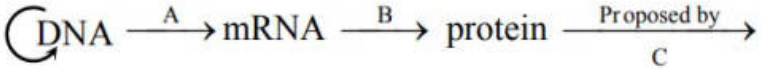
One of the most frequently used techniques in DNA fingerprinting is :-

(A) SSCP (B) SCAR (C) VNTR (D) AFLP

Open solution

Q120. NEET 2013

The diagram shows an important concept in the genetic implication of DNA. Fill in the blanks A to C.



(A) A-translation B-extension C-Rosalind Franklin

(B) A-translation B-transcription C-Erwin Chargaff

(C) A-transcription B-translation C-Francis Crick

(D) A-transcription B-replication C-James Watson

Open solution

Q121. NEET 2013

Which enzyme will be produced in a cell in which there is a non-sense mutation in the lac Y gene ?

(A) Transacetylase (B) Lactose permease and transacetylase

(C) Lactose permease (D) β-galactosidase

Open solution

Q122. AIPMT 2012 Mains

Read the following four statements (A-D) :
(A) In transcription, adenosine pairs with uracil
(B) Regulation of lac operon by repressor is referred to as positive regulation
(C) The human genome has approximately 50,000 genes
(D) Haemophilia is a sex-linked recessive disease
How many of the above statements are right ?

(A) Four (B) Two (C) Three (D) One

Open solution

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Q123. AIPMT 2012 Mains

What is it that forms the basis of DNA Fingerprinting ?

(A) The relative proportions of purines and pyrimidines in DNA

(B) The relative amount of DNA in the ridges and grooves of the fingerprints

(C) The relative difference in the DNA occurrence in blood, skin and saliva

(D) Satellite DNA occurring as highly repeated short DNA segments

Open solution

Q124. AIPMT 2012 Prelims

Removal of introns and joining of exons in a defined order during transcription is called

(A) Looping (B) Inducing (C) Slicing (D) Splicing

Open solution

Q125. AIPMT 2012 Prelims

If one strand of DNA has the nitrogenous base sequence as ATCTG, what would be the complementary RNA stand sequence

(A) ATCGU (B) UAGAC (C) AACTG (D) TTAGU

Open solution

Q126. AIPMT 2012 Prelims

Ribosomal RNA is actively synthesized in

(A) Nucleolus (B) Ribosomes (C) Lysosomes (D) Nucleoplasm

Open solution

Q127. AIPMT 2012 Prelims

Which one of the following is not a part of a transcription unit in DNA ?

(A) The structural gene (B) the inducer

(C) A terminator (D) A promoter

Open solution

Q128. AIPMT 2012 Prelims

Removal of RNA polymerase -III from nucleoplasm will affect the synthesis of

(A) tRNA (B) hnRNA (C) rRNA (D) mRNA

Open solution

Q129. AIPMT 2011 Mains

The unequivocal proof of DNA as the genetic material came from the studies on a :

(A) Viroid (B) Bacterium (C) Fungus (D) Bacterial virus

Open solution

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Q130. AIPMT 2011 Mains

In history of biology, human genome project led to the development of

(A) Bioinformatics (B) Biosystematics (C) Biotechnology (D) Biomonitoring

Open solution

Q131. AIPMT 2011 Prelims

Which one of the following also acts as a catalyst in a bacterial cell?

(A) 5S *r*RNA (B) snRNA (C) hnRNA (D) 23S *r*RNA

Open solution

Q132. AIPMT 2011 Prelims

What are those structures that appear as 'beads-on-string' in the chromosomes when viewed under electron microscope ?

(A) Nucleosomes (B) Genes (C) Nucleotides (D) Base pairs

Open solution

Q133. AIPMT 2010 Mains

The 3'-5' phosphodiester linkages inside a polynucleotide chain serve to join -

(A) One nucleotide with another nucleotide
(B) One nitrogenous base with pentose sugar
(C) One nucleoside with another nucleoside
(D) One DNA strand with the other DNA strand

Open solution

Q134. AIPMT 2010 Mains

The lac Operon consists of -

(A) Three regulatory genes and three structure genes
(B) Four regulatory genes only
(C) Two regulatory genes and two structural genes
(D) One regulatory gene and three structural genes

Open solution

Q135. AIPMT 2010 Mains

In eukaryotic cell transcription, RNA splicing and RNA capping take place inside the -

(A) Dictyosomes (B) Ribosomes (C) Nucleus (D) ER

Open solution

Q136. AIPMT 2010 Mains

Which one of the following statements about the particular entity is true ?

(A) Nucleosome is formed of nucleotides
(B) Centromere is found in animal cells, which produces aster during cell division
(C) The gene for producing insulin is present in every body cell
(D) DNA consists of a core of eight histones

Open solution

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Q137. AIPMT 2010 Prelims

Which one of the following does not follow the central dogma of molecular biology ?

(A) Mucor (B) HIV (C) Chlamydomonas (D) Pea

Open solution

Q138. AIPMT 2010 Prelims

The one aspect which is not a salient feature of genetic code, is its being –

(A) Specific (B) Degenerate (C) Ambiguous (D) Universal

Open solution

Q139. AIPMT 2010 Prelims

Select the two correct statements out of the four (a-d) statements given below about lac operon.
(a) Glucose or galactose may bind with the repressor and inactivate it
(b) In the absence of lactose the repressor binds with the operator region
(c) The z-gene codes for permease
(d) This was elucidated by Francois Jacob and Jacque Monod
The correct statements are :

(A) (b) and (d) (B) (a) and (b) (C) (b) and (c) (D) (a) and (c)

Open solution

Q140. AIPMT 2010 Prelims

Satellite DNA is useful tool in

(A) Genetic engineering (B) Organ transplantation
(C) Sex determination (D) Forensic science

Open solution

Q141. AIPMT 2009

Whose experiments cracked the DNA and discovered unequivocally that a genetic code is a "triplet" ?

(A) Nirenberg and Mathaei (B) Morgan and Sturtevant
(C) Hershey and Chase (D) Beadle and Tatum

Open solution

Q142. AIPMT 2009

Semi-conservative replication of DNA was first demonstrated in :

(A) Drosophila melanogaster (B) Streptococcus pneumoniae
(C) Salmonella typhimurium (D) Escherichia coli

Open solution

Q143. AIPMT 2009

Removal of introns and joining the exons in a defined order in a transcription unit is called :

(A) Transformation (B) Capping (C) Splicing (D) Tailing

Open solution

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Q144. AIPMT 2009

What is not true for genetic code?

(A) It is nearly universal

(B) A codon in mRNA is read in a noncontiguous fashion

(C) It is degenerate

(D) It is unambiguous

Open solution

Q145. AIPMT 2008

In the DNA molecule:

(A) the total amount of purine nucleotides and pyrimidine nucleotides is not always equal

(B) there are two strands which run antiparallel one in 5'→ 3' direction and other in 3'→ 5'

(C) there are two strands which run parallel in the 5'→ 3' direction

(D) the proportion of adenine in relation to thymine varies with the organism

Open solution

Q146. AIPMT 2008

Which one of the following pairs of nitrogenous bases of nucleic acids, is wrongly matched with the category mentioned against it?

(A) Thymine, Uracil — Pyrimidines

(B) Guanine, Adenine — Purines

(C) Uracil, Cytosine — Pyrimidines

(D) Adenine, Thymine — Purines

Open solution

Q147. AIPMT 2008

Polysome is formed by

(A) several ribosomes attached to a single mRNA

(B) many ribosomes attached to a strand of endoplasmic reticulum

(C) a ribosome with several subunits

(D) ribosomes attached to each other in a linear arrangement

Open solution

Q148. AIPMT 2008

Which one of the following pairs of codons is correctly matched with their function or the signal for the particular amino acid ?

(A) UUA, UCA — Leucine

(B) AUG, ACG — Start/Methionine

(C) UAG, UGA — Stop

(D) GUU, GCU — Alanine

Open solution

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Q149. AIPMT 2007

The Okazaki fragments in DNA chain growth

(A) polymerize in the 3'-to-5' direction and form the replication fork

(B) prove the semi-conservative nature of DNA replication

(C) polymerize in the 5'-to-3' direction and explain 3'-to-5' DNA replication

(D) result in transcription

Open solution

Q150. AIPMT 2007

The length of DNA molecule greatly exceeds the dimensions of the nucleus in eukaryotic cells. How is this DNA accommodated?

(A) Super-coiling in nucleosomes

(B) DNase digestion

(C) Through elimination of repetitive DNA

(D) Deletion of non-essential genes

Open solution

Q151. AIPMT 2007

During transcription, RNA polymerase holoenzyme binds to a gene promoter and assumes a saddle-like structure. What is its DNA-binding sequence?

(A) AATT (B) CACC (C) TATA (D) TTAA

Open solution

Q152. AIPMT 2007

Differentiation of organs and tissues in a developing organism is associated with

(A) differential expression of genes

(B) lethal mutations

(C) deletion of genes

(D) developmental mutations

Open solution

Q153. AIPMT 2007

One gene-one enzyme relationship was established for the first time in

(A) *Salmonella typhimurium*

(B) *Escherichia coli*

(C) *Diplococcus pneumoniae*

(D) *Neurospora crassa*

Open solution

Q154. AIPMT 2007

Telomere repetitive DNA sequences control the function of eukaryotic chromosomes because they

(A) are RNA transcription initiator

(B) help chromosome pairing

(C) prevent chromosome loss

(D) act as replicons

Open solution

Q155. AIPMT 2007

Molecular basis of organ differentiation depends on the modulation in transcription by

(A) ribosome

(B) transcription factor

(C) anticodon

(D) RNA polymerase

Open solution

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Q156. AIPMT 2007

The two polynucleotide chains in DNA are

(A) discontinuous (B) antiparallel (C) semiconservative (D) parallel

Open solution

Q157. AIPMT 2006

One turn of the helix in a B-form DNA is approximately -

(A) 20 nm (B) 3.4 nm (C) 2 nm (D) 0.34 nm

Open solution

Q158. AIPMT 2006

One gene-one enzyme hypothesis was postulated by -

(A) A.Garrod (B) Hershey and Chase
(C) Beadle and Tatum (D) R. Franklin

Open solution

Q159. AIPMT 2006

Antiparallel strands of a DNA molecule means that-

(A) The phosphate groups at the start of two DNA strands are in opposite position (pole)
(B) One strand turns anti-clockwise
(C) One strand turns clockwise
(D) The phosphate groups of two DNA strands, at their ends, share the same position

Open solution

Q160. AIPMT 2006

Which antibiotic inhibits interaction between tRNA and mRNA during bacterial protein synthesis ?

(A) Neomycin (B) Erythromycin (C) Tetracycline (D) Streptomycin

Open solution

Q161. AIPMT 2006

Amino acid sequence, in protein synthesis is decided by the sequence of -

(A) cDNA (B) mRNA (C) tRNA (D) rRNA

Open solution

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Q162. AIPMT 2005

E. coli cells with a mutated z gene of the lac operon cannot grow in medium containing only lactose as the source of energy because -

(A) They cannot transport lactose from the medium into the cell

(B) They cannot synthesize functional beta galactosidase

(C) In the presence of glucose, E. coli cells do not utilize lactose

(D) The lac operon is constitutively active in these cells

Open solution

Q163. AIPMT 2005

Telomerase is an enzyme which is a -

(A) RNA

(B) Repetitive DNA

(C) Simple protein

(D) Ribonucleoprotein

Open solution

Q164. AIPMT 2005

Using imprints from a plate with complete medium and carrying bacterial colonies, you can select streptomycin resistant mutants and prove that such mutations do not originates as adaptation. These imprints need to be used -

(A) On plates with minimal medium

(B) Only on plates with streptomycin

(C) Only on plates without streptomycin

(D) On plates with and without streptomycin

Open solution

Q165. AIPMT 2005

Protein synthesis in an animal cell occurs -

(A) On ribosomes present in the nucleolus as well as in cytoplasm

(B) On ribosomes presents in cytoplasm as well as in mitochondria

(C) Only on the ribosomes present in cytosol

(D) Only on ribosomes attached to the nuclear envelope and endoplasmic reticulum

Open solution

Q166. AIPMT 2005

Which one of the following makes use of RNA as a template to synthesize DNA -

(A) Reverse transcriptase

(B) DNA dependant RNA polymerase

(C) RNA polymerase

(D) DNA polymerase

Open solution

Q167. AIPMT 2005

During transcription holoenzyme RNA polymerase binds to a DNA sequence and the DNA assumes a saddle like structure at the point. What is the sequence called ?

(A) TATA box

(B) GGTT box

(C) CAAT box

(D) AAAT box

Open solution

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Q168. AIPMT 2004

During replication of a bacterial chromosome DNA synthesis starts from a replication origin site and

(A) RNA primers are involved
(B) is facilitated by telomerase
(C) moves in one direction of the site
(D) moves in bi-directional way.

Open solution

Q169. AIPMT 2004

After a mutation at a genetic locus the character of an organism changes due to the change in :-

(A) Protein synthesis pattern
(B) RNA transcription pattern
(C) DNA replication
(D) Protein structure

Open solution

Q170. AIPMT 2004

Which form of RNA has a structure resembling clover leaf ?

(A) r-RNA (B) m-RNA (C) t-RNA (D) hn-RNA

Open solution

Q171. AIPMT 2004

During transcription, if the nucleotide sequence of the DNA strand that is being coded is ATACG, then the nucleotide sequence in the mRNA would be -

(A) UAUGC (B) TCTGG (C) UATGC (D) TATGC

Open solution

Q172. AIPMT 2004

In a mutational event, when adenine is replaced by guanine, it is a case of -

(A) Frameshift mutation
(B) Transition
(C) Transversion
(D) Transcription

Open solution

Q173. AIPMT 2004

The following ratio is generally constant for a given species :-

(A) A + C / T + G (B) G + C / A+ T (C) A + G / C +T (D) T + C / G + A

Open solution

Q174. AIPMT 2004

DNA fingerprinting refers to :-

(A) Techniques used for molecular analysis of different specimens of DNA
(B) Anlysis of DNA samples using imprinting devices
(C) Molecular analysis of profiles of DNA samples
(D) Techniques used for identification of fingerprints of individuals

Open solution

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Q175. AIPMT 2003

What would happen if in a gene encoding a polypeptide of 50 amino acids, 25th codon (UAU) is mutated to UAA ?

(A) A polypeptide of 49 amino acids will be formed
(B) A polypeptide of 25 amino acids will be formed
(C) Two polypeptides of 24 and 25 amino acids will be formed
(D) A polypeptide of 24 amino acids will be formed

Open solution

Q176. AIPMT 2003

What does "lac" refer to in what we call the lac operon ?

(A) The number 1,00,000
(B) Lac insect
(C) Lactase
(D) Lactose

Open solution

Q177. AIPMT 2003

During translation initiation in prokaryotes, a GTP molecule is needed in : -

(A) Association of 30S mRNA with formy-lmet-tRNA
(B) Formation of formyl-met-tRNA
(C) Association of 50S subunit of ribosome with initiation complex
(D) Binding of 30S subunit of ribosome with mRNA

Open solution

Q178. AIPMT 2003

Which one of the following triplet codes, is correctly matched with its specificity for an amino acid in protein synthesis or as 'start' or 'stop' codon ?

(A) UUU – Stop
(B) UAC – Tyrosine
(C) UGU – Leucine
(D) UCG – Start

Open solution

Q179. AIPMT 2003

During transcription, the DNA site at which RNA polymerase binds is called :-

(A) Receptor
(B) Regulator
(C) Enhancer
(D) Promoter

Open solution

Q180. AIPMT 2003

Degeneration of a genetic code is attributed to the : -

(A) Second member of a codon
(B) Third member of a codon
(C) First member of a codon
(D) Entire codon

Open solution

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Q181. AIPMT 2003

In the genetic code dictionary, how many codons are used to code for all the 20 essential amino acids : -

(A) 64 (B) 20 (C) 60 (D) 61

Open solution

Q182. AIPMT 2002

In a DNA percentage of thymine is 20% then what is the percentage of guanine : -

(A) 40% (B) 20% (C) 30% (D) 60%

Open solution

Q183. AIPMT 2002

Transformation experiment was first performed on which bacteria : -

(A) Diplococcus pneumoniae (B) Pasteurella pestis

(C) E. coli (D) Salmonella

Open solution

Q184. AIPMT 2002

In E. Coli, during lactose metabolism repressor binds to : -

(A) Promoter gene (B) Operator gene (C) Structural gene (D) Regulator gene

Open solution

Q185. AIPMT 2002

Out of 64 codons, 61 codons code for 20 types of amino acid it is called : -

(A) Wobbling of codon (B) Universility of codons

(C) Overlapping of gene (D) Degeneracy of genetic code

Open solution

Q186. AIPMT 2002

Which of the following reunites the exon segments after RNA splicing ?

(A) RNA polymerase (B) RNA proteases (C) RNA primase (D) RNA ligase

Open solution

Q187. AIPMT 2002

Which of the following enzymes are used to join bits of DNA ?

(A) Endonuclease (B) DNA polymerase (C) Ligase (D) Primase

Open solution

Q188. AIPMT 2002

Exon part of m-RNAs have code for : -

(A) Lipid (B) Protein (C) Phospholipid (D) Carbohydrate

Open solution

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Q189. AIPMT 2002

Change in sequence of nucleotide in DNA is called as -

(A) Recombination (B) Mutation (C) Mutagen (D) Translation

Open solution

Q190. AIPMT 2001

m-RNA is synthesised on DNA template in which direction : -

(A) Any (B) Both (c) and (d) (C) 5' → 3' (D) 3' → 5'

Open solution

Q191. AIPMT 2001

In Negative operon : -

(A) cAMP has negative effect on lac operon

(B) Co-repressor binds with inducer

(C) Co-repressor binds with repressor

(D) Co-repressor does not bind with repressor

Open solution

Q192. AIPMT 2001

Gene and cistron words are sometimes used synonymously because : -

(A) One gene contains one cistron (B) One cistron contains many gene

(C) One gene contains no cistron (D) One gene contains many cistrons

Open solution

Q193. AIPMT 2001

Types of RNA polymerase required in nucleus for RNA synthesis : -

(A) 2 (B) 3 (C) 1 (D) 4

Open solution

Q194. AIPMT 2001

Due to discovery of which of the following in 1980's the evolution was termed as RNA world?

(A) m-RNA, t-RNA, r-RNA synthesize proteins (B) In some viruses, RNA is genetic material

(C) Some RNAs have enzymatic property (D) RNA is not found in all cells

Open solution

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Q195. AIPMT 2001

E. coli about to replicate was placed in a medium containing radioactive thymidine for five minutes. Then it was made to replicate in a normal medium. Which of the following observation shall be correct?

(A) Both the strands of DNA will be radioactive

(B) One strand radioactive

(C) Each strand half radioactive

(D) None is radioactive

Open solution

Q196. AIPMT 2001

At the time of organogenesis, genes regulate the process at different levels and at different time due to

(A) promoter (B) regulator (C) intron (D) exon

Open solution

Q197. AIPMT 2000

Method of DNA replication in which two strands of DNA separates and synthesize new strands

(A) Conservative (B) Non conservative (C) Dispersive (D) Semiconservative

Open solution

Q198. AIPMT 2000

Which of the following is initiation codon ?

(A) AUC (B) CCU (C) UAG (D) AUG

Open solution

Q199. AIPMT 2000

Anticodon occurs in :

(A) DNA (B) r-RNA (C) t-RNA (D) m-RNA

Open solution

Q200. AIPMT 2000

Length of one loop of B- DNA :

(A) 20 nm. (B) 10 nm. (C) 3.4 nm. (D) 0.34 nm.

Open solution

Q201. AIPMT 2000

In three dimensional view the molecule of t-RNA is :

(A) E-shaped (B) L-shaped (C) S-shaped (D) Y-shaped

Open solution

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Q202. AIPMT 2000

Similarity in DNA and RNA :

(A) Both have similar sugar
(B) Both are genetic material
(C) Both have similar pyrimidine
(D) Both are polymer of nucleotides

Open solution

Q203. AIPMT 2000

During organ differentiation in *Drosophila*, an organ is modified to another organ (such as wings may be replaced by legs). Genes responsible for such metamorphosis are called

(A) double dominant genes
(B) plastid genes
(C) complementary genes
(D) homeotic genes

Open solution

Q204. AIPMT 1999

The *Pneumococcus* experiment proves that :

(A) DNA is the genetic material
(B) RNA sometime controls the production of DNA and proteins
(C) bacteria undergo binary fission
(D) bacteria do not reproduce sexually

Open solution

Q205. AIPMT 1999

In operon concept, regulator gene functions as :

(A) repressor
(B) inhibitor
(C) regulator
(D) all of these

Open solution

Q206. AIPMT 1999

Initiation codon in eukaryotes is

(A) AUG
(B) AGU
(C) UAG
(D) GAU

Open solution

Q207. AIPMT 1999

DNA is mainly found in :

(A) nucleus only
(B) cytoplasm only
(C) nucleus and cytoplasm
(D) nucleolus

Open solution

Q208. AIPMT 1999

In DNA when AGCT occurs, their association is as per which of the following pair ?

(A) AC-GT
(B) AG-CT
(C) AT-GC
(D) All of these

Open solution

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Q209. AIPMT 1999

The eukaryotic genome differs from the prokaryotic genome because

(A) DNA is complexed with histones in prokaryotes

(B) repetitive sequences are present in eukaryotes

(C) genes in the former cases are organised into operons

(D) DNA is circular and single stranded in prokaryotes

Open solution

Q210. AIPMT 1998

Which base is responsible for hot spots for spontaneous point mutations?

(A) Guanine (B) Adenine (C) 5-bromouracil (D) 5-methylcytosine

Open solution

Q211. AIPMT 1998

Genes that are involved in turning on or off the transcription of a set of structural genes are called

(A) polymorphic genes (B) operator genes

(C) redundant genes (D) regulatory genes

Open solution

Q212. AIPMT 1998

DNA elements, which can switch their position, are called

(A) exons (B) introns (C) cistrons (D) transposons

Open solution

Q213. AIPMT 1997

Genes are packaged into a bacterial chromosome by

(A) acidic protein (B) actin (C) histones (D) basic protein

Open solution

Q214. AIPMT 1997

The codons causing chain termination are

(A) AGT, TAG, UGA (B) UAG, UGA, UAA (C) TAG, TAA, TGA (D) GAT, AAT, AGT

Open solution

Q215. AIPMT 1997

DNA synthesis can be specifically measured by estimating the incorporation of radio-labelled

(A) thymidine (B) deoxyribose sugar

(C) uracil (D) adenine

Open solution

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Q216. AIPMT 1997

The RNA that pick up specific amino acid from amino acid pool in the cytoplasm to ribosome during protein synthesis is called

(A) *r*RNA (B) RNA (C) *m*RNA (D) *t*RNA

Open solution

Q217. AIPMT 1997

Which of the following step of translation does not consume a high energy phosphate bond?

(A) Peptidyl transferase reaction (B) Aminoacyl *t*RNA binding to A-site
(C) Translocation (D) Amino acid activation

Open solution

Q218. AIPMT 1996

Which of the following serves as a terminal codon?

(A) UAG (B) AGA (C) AUG (D) GCG

Open solution

Q219. AIPMT 1996

The maximum formation of *m*RNA occurs in

(A) ribosome (B) nucleoplasm (C) cytoplasm (D) nucleolus

Open solution

Q220. AIPMT 1996

Radio-tracer technique shows that DNA is in

(A) multi-helix stage (B) single-helix stage
(C) double-helix stage (D) none of these

Open solution

Q221. AIPMT 1995

The wild type *E.coli* cells are growing in normal medium with glucose. They are transferred to a medium containing only lactose as sugar. Which of the following changes take place?

(A) The *lac* operon is induced. (B) *E.coli* cells stop dividing.
(C) The *lac* operon is repressed. (D) All operons are induced.

Open solution

Q222. AIPMT 1995

The *lac* operon is an example of

(A) repressible operon (B) overlapping genes
(C) arabinose operon (D) inducible operon

Open solution

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Q223. AIPMT 1995

An environmental agent, which triggers transcription from an operon, is a

(A) depressor
(B) controlling element
(C) regulator
(D) inducer

Open solution

Q224. AIPMT 1995

If the sequence of bases in DNA is ATTCGATG, then the sequence of bases in its transcript will be

(A) GUAGCUUA
(B) AUUCGAUG
(C) CAUCGAAU
(D) UAAGCUAC

Open solution

Q225. AIPMT 1995

If the DNA codons are ATG ATG ATG and a cytosine base is inserted at the beginning, then which of the following will result?

(A) CAT GAT GATG
(B) A non-sense mutation
(C) C ATG ATG ATG
(D) CA TGA TGA TG

Open solution

Q226. AIPMT 1995

In split genes, the coding sequences are called

(A) exons
(B) cistrons
(C) introns
(D) operons

Open solution

Q227. AIPMT 1995

Anticodon is an unpaired triplet of bases in an exposed position of

(A) *t*RNA
(B) *m*RNA
(C) *r*RNA
(D) both (b) and (c)

Open solution

Q228. AIPMT 1994

Initiation codon in eukaryotes is

(A) GAU
(B) AGU
(C) AUG
(D) UAG

Open solution

Q229. AIPMT 1994

'*Lac* operon' in *E. coli*, is induced by

(A) 'I' gene
(B) promoter gene
(C) β-galactosidase
(D) lactose

Open solution

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Q230. AIPMT 1994

There are special proteins that help to open up DNA double helix in front of the replication fork. These proteins are

(A) DNA ligase
(B) DNA topoisomerase I
(C) DNA gyrase
(D) DNA polymerase I

Open solution

Q231. AIPMT 1994

In protein synthesis, the polymerization of amino acids involves three steps. Which one of the following is not involved in the polymerisation of protein?

(A) Termination
(B) Initiation
(C) Elongation
(D) Transcription

Open solution

Q232. AIPMT 1993

Which is wrong about nucleic acids?

(A) DNA is single stranded in some viruses.
(B) RNA is double stranded occasionally.
(C) Length of one helix is 45 Å in B-DNA.
(D) One turn of Z-DNA has 12 bases.

Open solution

Q233. AIPMT 1993

Binding of specific protein on regulatory DNA sequence can be studied by means of

(A) ultra centrifugation
(B) electron microscope
(C) light microscope
(D) X-rays crystallography

Open solution

Q234. AIPMT 1993

Nucleosome core is made of

(A) H_1, H_2A, H_2B and H_3
(B) H_1, H_2A, H_2B, H_4
(C) H_1, H_2A, H_2B, H_3 and H_4
(D) H_2A, H_2B, H_3 and H_4

Open solution

Q235. AIPMT 1993

Initiation codon of protein synthesis (in eukaryotes) is

(A) GUA
(B) GCA
(C) CCA
(D) AUG

Open solution

Q236. AIPMT 1993

The transforming principle of *Pneumococcus* as found out by Avery, MacLeod and McCarty was

(A) *m*RNA
(B) DNA
(C) protein
(D) polysaccharide

Open solution

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Q237. AIPMT 1993

Who proved that DNA is basic genetic material?

(A) Griffith
(B) Watson
(C) Boveri and Sutton
(D) Hershey and Chase

Open solution

Q238. AIPMT 1993

Because most of the amino acids are represented by more than one codon, the genetic code is

(A) overlapping
(B) wobbling
(C) degenerate
(D) generate

Open solution

Q239. AIPMT 1993

During DNA replication, the strands separate by

(A) DNA polymerase
(B) topoisomerase
(C) unwindase/helicase
(D) gyrase

Open solution

Q240. AIPMT 1993

The process of translation is

(A) ribosome synthesis
(B) protein synthesis
(C) DNA synthesis
(D) RNA synthesis

Open solution

Q241. AIPMT 1993

A DNA with unequal nitrogen bases would most probably be

(A) single stranded
(B) double stranded
(C) triple stranded
(D) four stranded

Open solution

Q242. AIPMT 1993

Nucleotide arrangement in DNA can be seen by

(A) X-ray crystallography
(B) electron microscope
(C) ultracentrifuge
(D) light microscope

Open solution

Q243. AIPMT 1992

Experimental material in the study of DNA replication has been

(A) *Escherichia coli*
(B) *Neurospora crassa*
(C) *Pneumococcus*
(D) *Drosophila melanogaster*

Open solution

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Q244. AIPMT 1992

Khorana first deciphered the triplet codons of

(A) serine and isoleucine
(B) cysteine and valine
(C) tyrosine and tryptophan
(D) phenylalanine and methinonine

Open solution

Q245. AIPMT 1992

Escherichia coli fully labelled with ^{15}N is allowed to grow in ^{14}N medium. The two strands of DNA molecule of the first generation bacteria have

(A) different density and do not resemble parent DNA
(B) different density but resemble parent DNA
(C) same density and resemble parent DNA
(D) same density but do not resemble parent DNA

Open solution

Q246. AIPMT 1991

A segment of DNA has 120 adenine and 120 cytosine bases. The total number of nucleotides present in the segment is

(A) 120
(B) 240
(C) 60
(D) 480

Open solution

Q247. AIPMT 1991

The process of transfer of genetic information from DNA to RNA/formation of RNA from DNA

(A) transversion
(B) transcription
(C) translation
(D) translocation

Open solution

Q248. AIPMT 1990

Which is not consistent with double helical structure of DNA?

(A) A = T, C = G
(B) Density of DNA decreases on heating.
(C) A + T/C + G is not constant.
(D) Both (a) and (b)

Open solution

Q249. AIPMT 1990

An octamer of 4 histones complexed with DNA forms

(A) endosome
(B) nucleosome
(C) mesosome
(D) centromere

Open solution

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Q250. AIPMT 1990

In the genetic dictionary, there are 64 codons as

(A) 64 amino acids are to be coded

(B) 64 types of *t*RNAs are present

(C) there are 44 nonsense codons and 20 sense codons

(D) genetic code is triplet

Open solution

Q251. AIPMT 1989

DNA replication is

(A) conservative and discontinuous

(B) semi-conservative and semi-discontinuous

(C) semi-conservative and discontinuous

(D) conservative

Open solution

Q252. AIPMT 1988

In double helix of DNA, the two DNA strands are

(A) coiled around a common axis

(B) coiled around each other

(C) coiled differently

(D) coiled over protein sheath

Open solution

Q253. AIPMT 1988

Genetic code consists of

(A) adenine and guanine

(B) cytosine and uracil

(C) cytosine and guanine

(D) all the above

Open solution

MCQ (More than One Correct Answer) 2 questions

Q254. AIPMT 2014

Which one of the following is wrongly matched?

(A) Transcription – Writing information from DNA to t–RNA

(B) Operon – Structural genes, operator and promoter

(C) Repressor protein – Binds to operator to stop enzyme synthesis

(D) Translation – Using information in m–RNA to make protein.

Open solution

Q255. AIPMT 2002

Jacob and Monod studied lactose metabolism in E. coli and proposed operon concept. Operon concept applicable for :

(A) All prokaryotes and all eukaryotes

(B) All prokaryotes and some protozoans

(C) All prokaryotes

(D) All prokaryotes and some eukaryotes

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **C**	4. **D**	5. **C**
6. **A**	7. **A**	8. **A**	9. **A**	10. **D**
11. **C**	12. **B**	13. **C**	14. **C**	15. **A**
16. **D**	17. **A**	18. **C**	19. **A**	20. **C**
21. **D**	22. **D**	23. **B**	24. **A**	25. **D**
26. **B**	27. **C**	28. **C**	29. **B**	30. **C**
31. **B**	32. **B**	33. **D**	34. **A**	35. **A**
36. **D**	37. **A**	38. **D**	39. **C**	40. **B**
41. **B**	42. **B**	43. **C**	44. **D**	45. **D**
46. **A**	47. **C**	48. **A**	49. **C**	50. **B**
51. **A**	52. **A**	53. **C**	54. **C**	55. **D**
56. **C**	57. **C**	58. **C**	59. **D**	60. **A**
61. **C**	62. **C**	63. **A**	64. **A**	65. **C**
66. **B**	67. **B**	68. **D**	69. **C**	70. **D**
71. **B**	72. **A**	73. **D**	74. **D**	75. **B**
76. **D**	77. **C**	78. **D**	79. **C**	80. **B**
81. **A**	82. **A**	83. **B**	84. **B**	85. **D**
86. **A**	87. **A**	88. **A**	89. **C**	90. **B**
91. **D**	92. **D**	93. **C**	94. **C**	95. **A**
96. **C**	97. **A**	98. **A**	99. **B**	100. **B**
101. **C**	102. **A**	103. **A**	104. **C**	105. **C**
106. **D**	107. **C**	108. **C**	109. **A**	110. **A**
111. **B**	112. **A**	113. **C**	114. **C**	115. **C**
116. **A**	117. **D**	118. **D**	119. **C**	120. **C**
121. **D**	122. **B**	123. **D**	124. **D**	125. **B**
126. **A**	127. **B**	128. **A**	129. **D**	130. **A**
131. **D**	132. **A**	133. **A**	134. **D**	135. **C**
136. **C**	137. **B**	138. **C**	139. **A**	140. **D**
141. **A**	142. **D**	143. **C**	144. **B**	145. **B**
146. **D**	147. **A**	148. **C**	149. **C**	150. **A**

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151. **C**	152. **A**	153. **D**	154. **C**	155. **D**
156. **B**	157. **B**	158. **C**	159. **A**	160. **A**
161. **B**	162. **B**	163. **D**	164. **B**	165. **B**
166. **A**	167. **A**	168. **D**	169. **D**	170. **C**
171. **A**	172. **B**	173. **B**	174. **C**	175. **D**
176. **D**	177. **A**	178. **B**	179. **D**	180. **B**
181. **D**	182. **C**	183. **A**	184. **B**	185. **D**
186. **D**	187. **C**	188. **B**	189. **B**	190. **C**
191. **C**	192. **A**	193. **B**	194. **C**	195. **B**
196. **B**	197. **D**	198. **D**	199. **C**	200. **C**
201. **B**	202. **D**	203. **D**	204. **A**	205. **A**
206. **A**	207. **A**	208. **C**	209. **B**	210. **D**
211. **D**	212. **D**	213. **D**	214. **B**	215. **A**
216. **D**	217. **A**	218. **A**	219. **B**	220. **C**
221. **A**	222. **D**	223. **D**	224. **D**	225. **A**
226. **A**	227. **A**	228. **C**	229. **D**	230. **B**
231. **D**	232. **C**	233. **D**	234. **D**	235. **D**
236. **B**	237. **D**	238. **C**	239. **C**	240. **B**
241. **A**	242. **A**	243. **A**	244. **B**	245. **A**
246. **D**	247. **B**	248. **C**	249. **B**	250. **D**
251. **B**	252. **A**	253. **D**		

MCQ (More than One Correct Answer)

254. **A**	255. **C, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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